

Atmoce-Cloud

API Reference

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1 Overview

This document describes Atmoce-Cloud application programming interfaces (APIs) and how to obtain Atmoce-Cloud data by calling APIs, including system access authentication and API usage.

2 API Overview

2.1 Principles

- The API is RESTful and comply with the HTTP protocol.
- The API response format is JSON.

2.1.1 Request Methods

Supported HTTP request methods are GET, POST, PUT, and DELETE.

- GET: queries resource information.
- POST: creates resources.
- PUT: updates resources.
- DELETE: deletes resources.

2.1.2 HTTP Status Codes

Atmoce-Cloud returns HTTP status codes for API calls. The first digit in an HTTP status code indicates the type of the response.

- 1xx informational response: The request was received, and request processing continues.

- 2xx success: The request was successfully received, understood, and accepted by the server.
- 3xx redirection: Further action needs to be taken to complete the request.
- 4xx client error: The request contains a syntax error or cannot be fulfilled.
 - 400: The request parameters are invalid.
 - 401: The request is unauthorized. The API token is invalid or has expired.
 - 404: The requested resource cannot be found.
 - 405: The request method is not allowed.
 - 422: The request is invalid.
 - 429: The number of requests reaches the upper limit.
- 5xx server error: The server failed to fulfill the request.
 - 500: An internal server error has occurred.
 - 501: The request was not implemented.

2.1.3 Requests and Responses

Sample request parameters

```
Request url: https://xxxxxxxx/openapi/v1/sites/getSites&page=1
```

Sample success response

```
{
  "success": true,
  "code": 200,
  "reason": null,
```

```
"details": null,
"data": [
  {
    "siteId": "1",
    "name": "site 1",
    "buildState": "completed",
    "buildStateName": "",
    "countryGec": null,
    "countryStanag": null,
    "address": null,
    "zipCode": null,
    "timeZone": null,
    "constructionMethod": "housetop",
    "solarCapacity": 1234,
    "batteryCapacity": 1234,
    "ownerName": null,
    "ownerMail": null
  },
  {
    "siteId": "2",
    "name": "site 2",
    "buildState": "building",
    "countryGec": null,
    "countryStanag": null,
    "address": null,
    "zipCode": null,
    "timeZone": null,
    "constructionMethod": "balcony",
    "solarCapacity": 4321,
    "batteryCapacity": 4321,
    "ownerName": null,
    "ownerMail": null
  }
]
```

2.2 Security Guarantee

- **HTTPS encryption:** All APIs are called over the HTTPS protocol to ensure security during data transmission.

- **API token authentication:** All APIs are authenticated in sessions, which are generated by using API tokens.

[Note: Keep your API token properly and replace it regularly.](#)

- **Data masking:** Sensitive data in the returned content is masked, such as the email address lu***@gmail.com.

2.3 QPS Limits

To prevent malicious attacks and resource abuse, the following queries per second (QPS) limits apply to API calls:

1. Each API token can be used to make up to 10,000 calls per month.
2. Each API token can be used to make up to 5 calls at a time.

API callers must observe such limits and plan the API call frequency to avoid triggering QPS throttling. If the number of API calls exceeds a limit, HTTP status code 429 is returned.

[Note: QPS limits may change as the system evolves. For more information, see the latest API reference.](#)

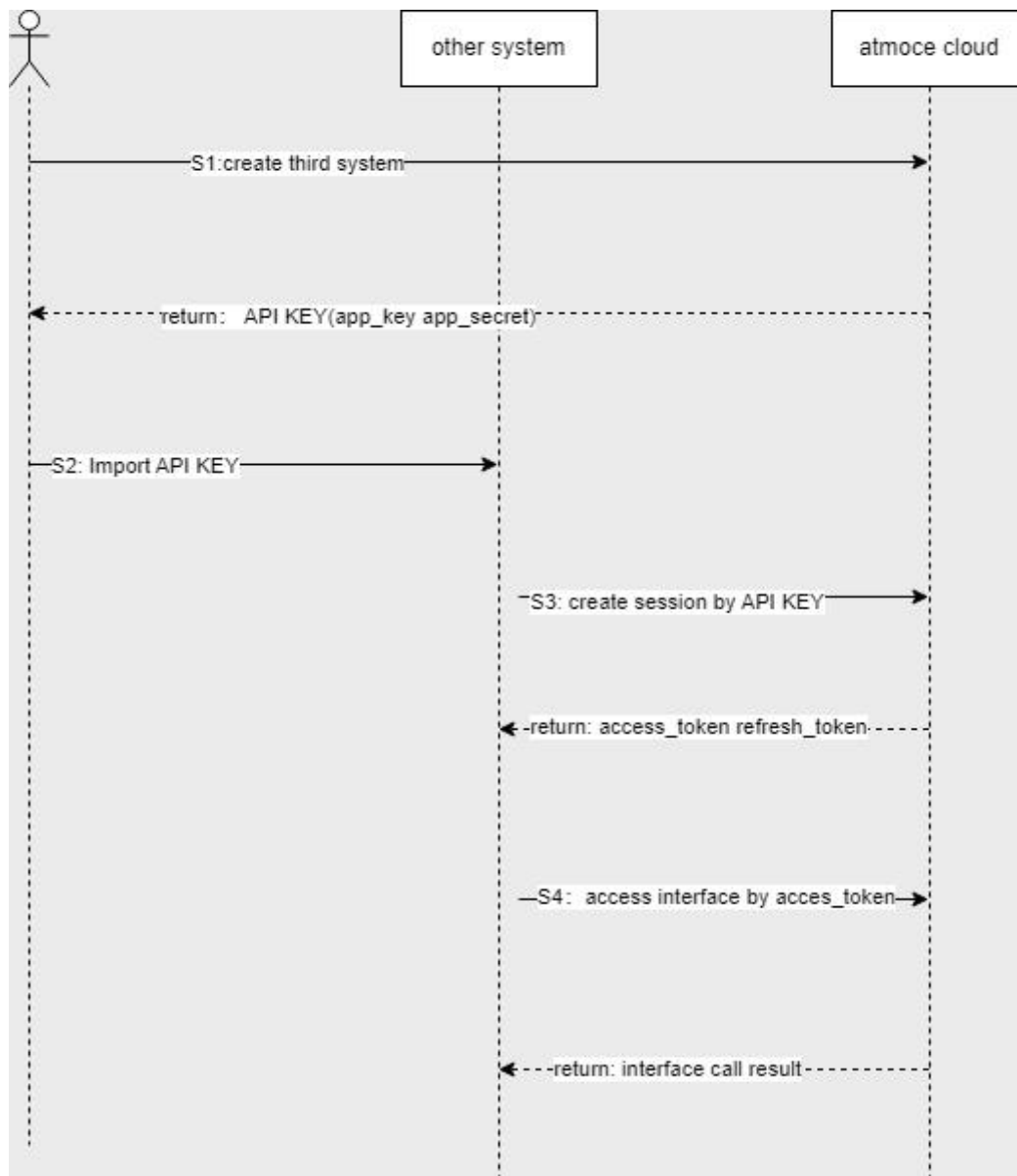
3 System Access Authentication

3.1 API Token

3.1.1 Procedure

Atmoce-Cloud authenticates API calls by using API tokens. The caller system needs to obtain a unique API token for authentication and obtain

the session required for an API call. The following figure shows the procedure.



- Steps 1 and 2: Atmoce-Cloud adds the caller system and grants the API call permissions to the caller system. Then, Atmoce-Cloud generates and sends an API token to the caller system.
- Step 3: The caller system uses the API token for authentication. Atmoce-Cloud generates and sends session information to the caller system. The session information includes a session token

(**access_token**) and a long-lived token (**refresh_token**).

- Step 4: The caller system carries the session token in the HTTP header to ensure a successful API call.

Note:

1. The session token is valid for 30 days. If the session token expires, it can be updated by using the long-lived token. The long-lived token is valid for 180 days. If the long-lived token expires, you need to generate another session by using the API token.
2. If an API call error occurs, the caller system needs to handle the error and record detailed call logs for troubleshooting.

3.1.2 Creating the API Token Required by the Caller System

Step 1: The installer enters the following information and sends the information to Atmoce-Cloud.

Item	Description	Required
<i>Atmoce-Cloud API License Agreement</i>	By sending an API request, the caller system acknowledges that they have read and agreed to <i>Atmoce-Cloud API License Agreement</i> .	Yes
Installer Name	Example: XXXX	Yes
System Name	Example: XXXX	Yes
System Overview	Example: Atmoce-Cloud supports the management of various energy resources, such as solar energy, battery storage, and EV chargers, providing customers with convenient energy services.	Yes

Step 2: Atmoce-Cloud assists in adding the caller system and sends the following information to the caller system.

Parameter	Type	Description
app_key	String	The API key.
app_secret	String	The API secret.

3.1.3 Generating and Closing a Session

3.1.3.1 Generating a Session

3.1.3.1.1 API Overview

Generates and updates a session token and a long-lived token for API calls. The session token is valid for 30 days. If the session token expires, it can be updated by using the long-lived token. The long-lived token is valid for 180 days. If the long-lived token expires, you need to generate another session by using the API token.

Note: If no session is generated or the session has expired for an API call, HTTP status code 401 is returned for the API call.

3.1.3.1.2 Request URL

`https://xxxxxxxxx/openapi/v1/auth/auth_token`

3.1.3.1.3 Request Method

HTTP method: POST

3.1.3.1.4 Request Parameters

Parameter	Position	Type	Required	Description
body	body	object	Yes	
» app_key	body	string	No	The API key for system authentication. This parameter is required for session generation.
» app_secret	body	string	No	The API secret for system authentication. This parameter is required for session generation.
» grant_type	body	string	Yes	system : generates a session. refresh : refreshes a session.
» refresh_token	body	string	No	The long-lived token. This parameter is required for session refreshing.

3.1.3.1.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	object		
»» access_token	string		The session token, which is contained in the header parameter Authorization in an API call.
»» refresh_token	string		The long-lived token, which is used to update an expired session token.

3.1.3.2 Closing a Session

3.1.3.2.1 API Overview

Closes a session. After a session is closed, both the session token and long-lived token become invalid.

Note: If a closed session is used for an API call, HTTP status code 401 is returned for the API call.

3.1.3.2.2 Request URL

`https://xxxxxxxxx/openapi/v1/auth/cancel_token`

3.1.3.2.3 Request Method

HTTP method: DELETE

3.1.3.2.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.

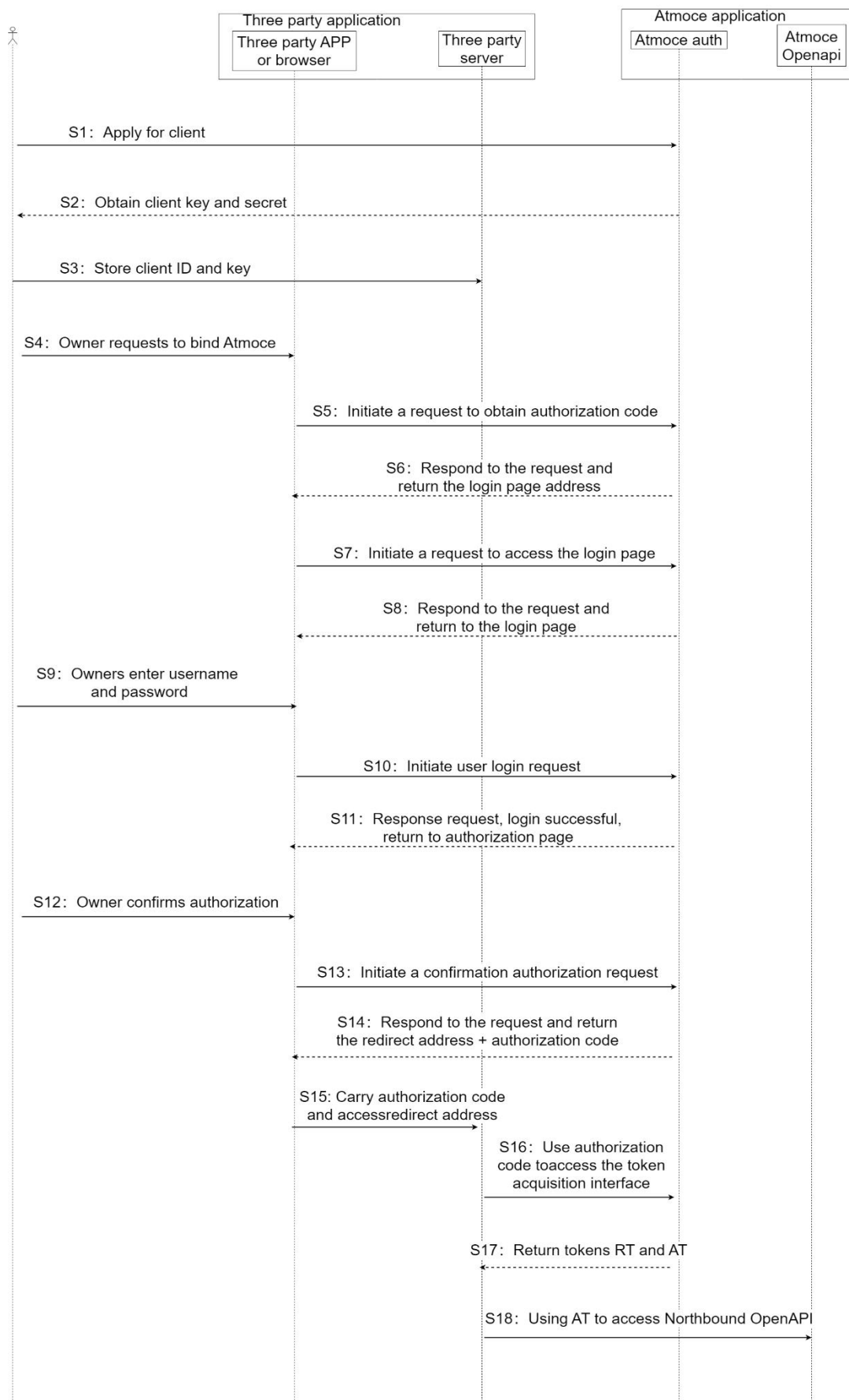
3.1.3.2.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		

3.2 OAuth 2.0

3.2.1 Procedure

Atmoce-Cloud can also authorize access from other systems by using OAuth 2.0. The caller system needs to obtain user authorization and the session required for an API call. The following figure shows the procedure.



- Steps 1 to 3: Atmoce-Cloud adds the caller system and generates the OAuth 2.0 client information (**OAuth_Client** and **OAuth_Secret**).
- Steps 4 to 14: The caller system integrates the HTML5 page of Atmoce-Cloud. A user logs in to grant the site and API call permissions. Atmoce-Cloud sends the authorization code to the callback URL.
- Steps 15 to 17: The caller system calls the `auth_token` API to generate a session based on the authorization code, client ID, and client key. **access_token** is the session token, and **refresh_token** is the long-lived token.
- Step 18: The caller system carries the session token in the HTTP header to ensure a successful API call.

Note:

1. The session token is valid for 30 days. If the session token expires, it can be updated by using the long-lived token. The long-lived token is valid for 180 days. If the long-lived token expires, you need to generate another session by using the API token.
2. If an API call error occurs, the caller system needs to handle the error and record detailed call logs for troubleshooting.

3.2.2 Creating the OAuth 2.0 Client

Step 1: The caller system provides the following information.

Item	Description	Required
<i>Atmoce-Cloud API License Agreement</i>	By sending an API request, the caller system acknowledges that they have read and agreed to <i>Atmoce-Cloud API License Agreement</i> .	Yes
Enterprise Name	Example: XXXX	Yes
Country/Region	Example: XXX	Yes
Email	Example: XXXX@gmail.com	Yes
System Name	Example: XXXX	Yes
图标	例如: 	Yes
System Overview	Example: Atmoce-Cloud supports the management of various energy resources, such as solar energy, battery storage, and EV chargers, providing customers with convenient energy services.	Yes
Callback URL	Example: https://www.example.com	Yes

Step 2: Atmoce-Cloud assists in creating the OAuth 2.0 client and sends the following information to the caller system.

Parameter	Type	Description
OAuth_Client	String	The client ID.
OAuth_Secret	String	The client key.

3.2.3 Authorizing the OAuth 2.0 Client

3.2.3.1 API Overview

Authorizes the OAuth 2.0 client in a caller system.

3.2.3.2 Request URL

<https://xxxxxxxxx/openapi/v1/auth/authorize>

3.2.3.3 Request Method

HTTP method: GET

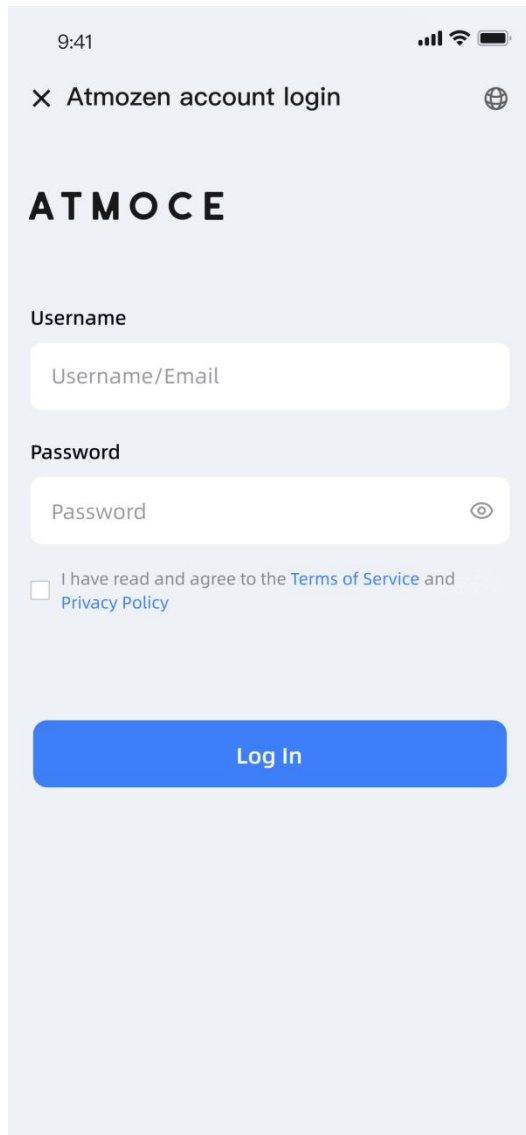
3.2.3.4 Request Parameters

Parameter	Position	Type	Required	Description
OAuth_Client	url	string	No	The client ID for system authentication. This parameter is required for session generation.
response_type	url	string	Yes	The type of the response. Default value: code .
redirect_uri	url	string	No	The callback URL, which must be consistent with that for creating the OAuth 2.0 client. Example: https://www.example.com

3.2.3.5 Response Parameters

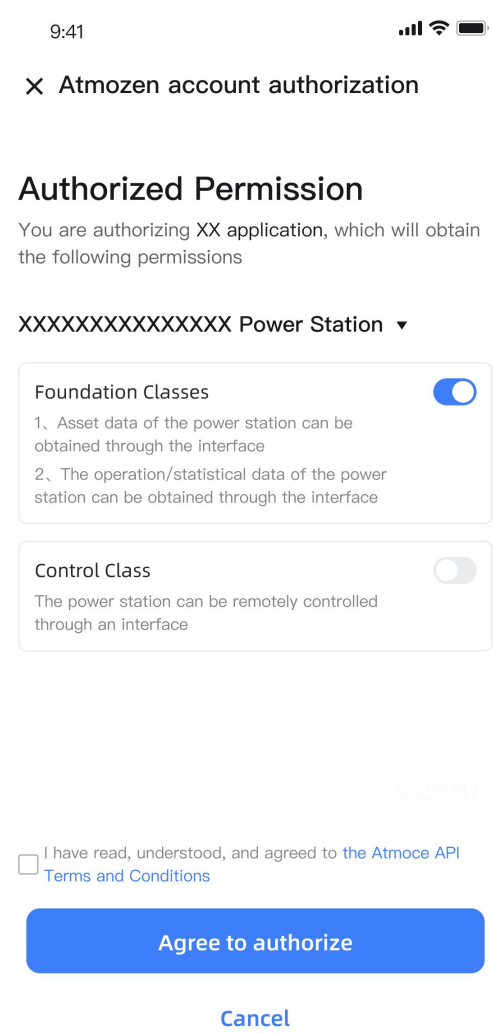
After Atmoce-Cloud receives the authorization request, authorization is performed in several steps.

Step 1: A user logs in.



The screenshot shows a mobile application interface for logging into an Atmozen account. At the top, the status bar displays the time 9:41, signal strength, Wi-Fi, and battery icons. Below the status bar, there is a header with a close icon (X), the text "Atmozen account login", and a globe icon. The main heading "ATMOCE" is prominently displayed. Underneath, there are two input fields: "Username" with a placeholder "Username/Email" and "Password" with a placeholder "Password" and a toggle icon (eye). Below the password field, there is a checkbox and the text "I have read and agree to the [Terms of Service](#) and [Privacy Policy](#)". At the bottom, there is a large blue button labeled "Log In".

Step 2: The user grants the site and API call permissions.



Step 3: Atmoce-Cloud sends the generated authorization code by using the **redirect_uri** parameter.

Parameter	Position	Type	Required	Description
Code	url	string	Yes	The authorization code that is used to generate an API session.

Example: <https://www.example.com/?Code=XXXXXXXXXXXX>

3.2.4 Generating and Closing a Session for the OAuth 2.0 Client

3.2.4.1 Generating a Session

3.2.4.1.1 API Overview

Generates and updates a session token and a long-lived token for API calls. The session token is valid for 30 days. If the session token expires, it can be updated by using the long-lived token. The long-lived token is valid for 180 days. If the long-lived token expires, you need to generate another session by using the API token.

Note: If no session is generated or the session has expired for an API call, HTTP status code 401 is returned for the API call.

3.2.4.1.2 Request URL

`https://xxxxxxxxx/openapi/v1/auth/auth_token`

3.2.4.1.3 Request Method

HTTP method: POST

3.2.4.1.4 Request Parameters

Parameter	Position	Type	Required	Description
body	body	object	Yes	
» grant_type	body	string	Yes	The authorization type. authorization_code : generates a session. refresh : refreshes a session.
» OAuth_Client	body	string	No	The client ID for system authentication. This parameter is required for session generation.
» OAuth_secret	body	string	No	The client key for system authentication. This parameter is required for session generation.
» refresh_token	body	string	No	The long-lived token. This parameter is required for session refreshing.
» Code	body	string	Yes	The authorization code.

» redirect_uri	body	string	Yes	The callback URL.
----------------	------	--------	-----	-------------------

3.2.4.1.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	object		
»» access_token	string		The session token, which is contained in the header parameter Authorization in an API call.
»» refresh_token	string		The long-lived token, which is used to update an expired session token.

3.2.4.2 Closing a Session

3.2.4.2.1 API Overview

Closes a session. After a session is closed, both the session token and long-lived token become invalid.

Note: If a closed session is used for an API call, HTTP status code 401 is returned for the API call.

3.2.4.2.2 Request URL

https://xxxxxxxx/openapi/v1/auth/cancel_token

3.2.4.2.3 Request Method

HTTP method: DELETE

3.2.4.2.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.

3.2.4.2.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		

4 Basic APIs

Basic APIs allow you to query basic information about sites and devices, such as the ID, name, serial number (SN), type, and model.

4.1 Querying a Site List

4.1.1 API Overview

Queries the list of all authorized sites. Up to 100 sites can be returned at a time. If the number of sites exceeds 100, another call is required, and pagination is used. If **pageEnd=1** is returned, all sites have been returned.

Note:

1. The site ID used by other APIs is the value of **siteId** returned by this API.
2. It is recommended to call this API once a day to obtain the information about new sites or changed sites.

4.1.2 Request URL

`https://xxxxxxxx/openapi/v1/sites/getSites`

4.1.3 Request Method

HTTP method: POST

4.1.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
page	url	integer	No	The page number. Default value: 1 .

4.1.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» pageEnd	Integer		The pagination status. 0 : incomplete. 1 : complete.
» data	list		The list of sites.
»»siteId	string		The site ID.
»»name	string		The site name.
»»buildState	string		The site provisioning status. Uncompleted : not provisioned. Completed : provisioned.
»»countryGEC	string		The Geopolitical Entities and Codes (GEC) country code.
»»countrySTANAG	string		The Standardization Agreement (STANAG) 1059 country code.
»»timeZone	string		The time zone. Example: UTC-04:00.
»»solarCapacity	double	kWp	The total component capacity.
»»MICapacity	double	kWp	The rated power of a microinverter.
»»batteryCapacity	double	kWh	The battery storage capacity.
»»gridTiedTime	string		The installation date. Example: dd/MM/yyyy.

4.2 Querying Basic Site Information

4.2.1 API Overview

Queries basic information about sites based on the site ID. Up to 100 sites can be queried at a time.

4.2.2 Request URL

<https://xxxxxxxxx/openapi/v1/sites/getSite>

4.2.3 Request Method

HTTP method: GET

4.2.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteIds	url	String	Yes	The site IDs. Up to 100 site IDs are supported.

4.2.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»»siteId	string		The site ID.
»»name	string		The site name.
»»buildState	string		The site provisioning status. uncompleted : not provisioned. completed : provisioned.
»»countryGEC	string		The GEC country code.
»»countrySTANAG	string		The STANAG 1059 country code.
»»zipCode	string		The postal code.
»»timeZone	string		The time zone.
»»solarCapacity	double	kWp	The total component capacity.

»»MICapacity	double	kWp	The rated power of a microinverter.
»»batteryCapacity	double	kWh	The battery storage capacity.
»»gridTiedTime	string		The installation date. Example: dd/MM/yyyy.

4.3 Querying a Site Device List

4.3.1 API Overview

Queries the list of all devices at sites based on the site ID. Up to 100 sites can be queried at a time.

Note: The device SN used by other APIs is the value of **deviceSN** returned by this API.

4.3.2 Request URL

<https://xxxxxxxx/openapi/v1/device/getDevicesBySite>

4.3.3 Request Method

HTTP method: GET

4.3.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteIds	url	String	Yes	The site IDs. Up to 100 site IDs are supported.

4.3.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true: The request was successful. false: The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see
» reason	string		
» details	string		

			Status Codes in the appendix.
» data	list		
»»siteId	string		The site ID.
»»deviceSN	string		The device SN.
»»deviceType	string		The device type. For more information, see Device Types in the appendix.
»»deviceMode	string		The device model.
»»MICapacity	double	W	The rated power of a microinverter.
»»batteryCapacity	double	kWh	The battery storage capacity.
»»batteryChargeMaxPower	double	kW	The maximum charging power of a battery.
»»batteryDisChargeMaxPower	double	kW	The maximum discharging power of a battery.

5 Monitoring APIs

Monitoring APIs allow you to query the operating data and statistics of sites and devices.

1. Operating data: The shortest collection interval is 15 minutes, and 96 data points are collected per day.
2. Statistics: Data is collected by day, month, or year.

5.1 Site

5.1.1 Querying the Latest Site Data

5.1.1.1 API Overview

Queries the latest site data. Up to 100 sites can be queried at a time. If the number of sites exceeds 100, another call is required.

5.1.1.2 Request URL

<https://xxxxxxxxx/openapi/v1/sites/getSitesLastPower>

5.1.1.3 Request Method

HTTP method: GET

5.1.1.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	String	Yes	The session token.
siteIds	url	String	Yes	The site IDs. Up to 100 site IDs are supported.

5.1.1.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» siteId	string		The site ID.
»» lastReportedTime	Long		The data update time, in milliseconds.
»» status	string		The site status. Uncompleted : inactivated. Normal : normal. Fault : faulty. Offline : offline.
»» solarGenerationPower	integer	W	The solar active power.
»» solarReactivePower	Integer	Var	The solar reactive power.
»» dailySoalrGeneration	double	kWh	The solar power daily produced.
»» monthlySoalrGeneration	double	kWh	The solar power monthly produced.
»» yearlySoalrGeneration	double	kWh	The solar power yearly produced.
»» lifetimeSoalrGeneration	double	kWh	The solar power cumulatively produced.
»» gridPower	integer	W	The grid power.
»» dailyGridExport	double	kWh	The power daily exported.
»» dailyGridImport	double	kWh	The power daily imported.
»» monthlyGridExport	double	kWh	The power monthly exported.
»» monthlyGridImport	double	kWh	The power monthly imported.
»» yearlyConsumption	double	kWh	The power yearly consumed.
»» yearlyGridExport	double	kWh	The power yearly exported.
»» lifetimeGridExport	double	kWh	The power cumulatively exported.

»» lifetimeGridImport	double	kWh	The power cumulatively imported.
»» consumptionPower	integer	W	The consumed power.
»» dailyConsumption	double	kWh	The power daily consumed.
»» monthlyConsumption	double	kWh	The power monthly consumed.
»» yearlyGridImport	double	kWh	The power yearly imported.
»» lifetimeConsumption	double	kWh	The power cumulatively consumed.
»» batteryPower	integer	W	The battery power.
»» batterySOC	double	%	The battery state of charge (SOC).
»» batteryStatus	string		The battery status. Charge : charging. discharge : discharging. normal : not charged or discharged.
»» batteryMode	string		The battery working mode. selfUse : self-powered. touMode : time of use (TOU).
»» dailyBatteryCharging	double	kWh	The battery power daily charged.
»» dailyBatteryDischarge	double	kWh	The battery power daily discharged.
»» monthly_batteryCharging	double	kWh	The battery power monthly charged.
»» monthlyBatteryCharging	double	kWh	The battery power monthly discharged.
»» yearlyBatteryCharging	double	kWh	The battery power yearly charged.
»» yearlyBatteryDischarge	double	kWh	The battery power yearly discharged.
»» lifetimeBatteryCharging	double	kWh	The battery power cumulatively charged.
»» lifetimeBatteryDischarge	double	kWh	The battery power cumulatively discharged.
»» totalCo2Reduced	double	kg	The CO ₂ cumulatively reduced.
»» totalTreesPlanted	double		The cumulatively equivalent tree planting.

5.1.2 Querying Site Energy

5.1.2.1 API Overview

Queries site power statistics. Up to 100 sites can be queried at a time. If the number of sites exceeds 100, another call is required.

Data can be queried by day, month, and year. By default, data of the current day is queried. To query data of another day, specify the date type

and date.

Limits: Data can be queried within 30 days or 12 months. If the time range to query exceeds a limit, status code 2007 is returned.

5.1.2.2 Request URL

`https://xxxxxxxxx/openapi/v1/sites/getSitesEnergy`

5.1.2.3 Request Method

HTTP method: GET

5.1.2.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	String	Yes	The session token.
siteIds	url	String	Yes	The site IDs. Up to 100 site IDs are supported.
dateType	url	String	No	The date type, such as day, month, or year. Default value: day.
beginDate	body	String	No	Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy. Default value: current day.
endDate	body	String	No	Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy. Default value: current day, which is the same as beginDate .

5.1.2.5 Response Parameters

Parameter	Type	Unit	Description
» success	Boolean		true : The request was successful. false : The request failed.
» code	Integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	String		
» details	String		
» data	list		

»» siteId	String		The site ID.
»» dateType	String		The date type, such as day, month, or year.
»» date	string		Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy.
»» soalrGeneration	Double	kWh	The solar power produced.
»» consumption	Double	kWh	The power consumed.
»» gridExport	Double	kWh	The power exported.
»» gridImport	Double	kWh	The power imported.
»» batteryCharging	Double	kWh	The battery power charged.
»» batteryDischarge	Double	kWh	The battery power discharged.

5.2 Microinverter

5.2.1 Querying the Latest Microinverter Data

5.2.1.1 API Overview

Queries the latest microinverter data based on the site ID or device SNs.

1. Site ID: Only one site can be queried at a time. All microinverters at the site are queried.
2. Device SNs: Up to 100 microinverters can be queried at a time.

Note:

1. The site ID and device SNs are mutually exclusive. For example, if the site ID is specified, device SNs are invalid.
2. A 1-in-1 microinverter does not contain branch data, and **null** is returned for branch data.

5.2.1.2 Request URL

<https://xxxxxxxx/openapi/v1/microInverter/getMIsLastData>

5.2.1.3 Request Method

HTTP method: GET

5.2.1.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteId	url	string	No	siteId and SNs are mutually exclusive. If siteId is specified, SNs is invalid. siteId : the site ID. Only one site ID is supported. SNs : the microinverter SNs. Up to 100 SNs are supported.
SNs	url	string	No	

5.2.1.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» siteId	string		The site ID.
»» SN	string		The microinverter SN.
»» lastReportedTime	Long		The data update time, in milliseconds.
»» status	string		The microinverter status. Offline : offline. onGrid : on-grid. standby : standby. offManual : powered off. offError : powered off unexpectedly. waitLight : waiting for light. upgrading : upgrading.
»» generationPower	Integer	W	The generated power.
»» dailyGeneration	Double	kWh	The power daily produced.
»» monthlyGeneration	Double	kWh	The power monthly produced.
»» yearlyGeneration	Double	kWh	The power yearly produced.
»» lifetimeGeneration	Double	kWh	The power cumulatively produced.
»» pvData	List		The branch data.

			This parameter is returned if more than two solar panels are queried.
»»»» pvNumber	Integer		The number of branches.
»»»» pvPower	Integer	V	The power of branch N.
»»»» pvDailyGeneration	Double	kWh	The power daily produced of branch N.
»»»»pvMonthlyGeneration	Double	kWh	The power monthly produced of branch N.
»»»» pvYearlyGeneration	Double	kWh	The power yearly produced of branch N.
»»»»pvLifetimeGeneration	Double	kWh	The power cumulatively produced of branch N.

5.2.2 Querying Historical Microinverter Data

5.2.2.1 API Overview

Queries historical microinverter data based on the site ID or device SNs.

By default, historical data of the current day is queried. To query data of another day, specify the date.

1. Site ID: Only one site can be queried at a time. All microinverters at the site are queried.
2. Device SNs: Up to 100 microinverters can be queried at a time.

Note:

1. The site ID and device SNs are mutually exclusive. If the site ID is specified, device SNs are invalid.
2. A 1-in-1 microinverter does not contain branch data, and **null** is returned for branch data.

5.2.2.2 Request URL

<https://xxxxxxxx/openapi/v1/microInverter/getMIsHistoryData>

5.2.2.3 Request Method

HTTP method: GET

5.2.2.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteId	url	string	No	siteId and SNs are mutually exclusive. If siteId is specified, SNs is invalid. siteId : the site ID. Only one site ID is supported. SNs : the microinverter SNs. Up to 100 SNs are supported.
SNs	url	string	No	
date	url	string	No	Format: dd/MM/yyyy. Default value: current day.

5.2.2.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» siteId	string		The site ID.
»» SN	string		The microinverter SN.
»» dateTime	String		Format: dd/MM/yyyy HH: mm. Interval: 15 minutes.
»» generationPower	Integer	W	The generated power.
»» pvData	List		The branch data. This parameter is returned if more than two solar panels are queried.
»»» pvNumber	Integer		The number of branches.
»»» pvPower	Integer	V	The power of branch N.

5.2.3 Querying the Produced Power of Microinverters

5.2.3.1 API Overview

Queries the historical power data of microinverters based on the site ID or device SNs. Data can be queried by day, month, and year. By default, data of the current day is queried. To query data of another day, specify the date type and date.

1. Site ID: Only one site can be queried at a time. All microinverters at the site are queried.
2. Device SNs: Up to 100 microinverters can be queried at a time.

Limits: Data can be queried within 30 days or 12 months. If the time range to query exceeds a limit, status code 2007 is returned.

Note:

1. The site ID and device SNs are mutually exclusive. If the site ID is specified, device SNs are invalid.
2. A 1-in-1 microinverter does not contain branch data, and **null** is returned for branch data.

5.2.3.2 Request URL

`https://xxxxxxxx/openapi/v1/microInverter/getMIsDayData`

5.2.3.3 Request Method

HTTP method: GET

5.2.3.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteId	url	string	No	siteId and SNs are mutually exclusive. If siteId is specified, SNs is invalid. siteId : the site ID. Only one site ID is
SNs	url	string	No	

				supported. SNs : the microinverter SNs. Up to 100 SNs are supported.
dateType	url	string	No	The date type, such as day, month, or year. Default value: day.
beginDate	body	string	No	Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy. Default value: current day.
endDate	body	string	No	Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy. Default value: current day, which is the same as beginDate .

5.2.3.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» siteId	string		The site ID.
»» SN	string		The microinverter SN.
»» dateType	string		The date type, such as day, month, or year.
»» date	string		Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy.
»» generation	Double	kWh	The power produced.
»» PVDayDataCO	List		The power produced of a branch.
»» pvNumber	Integer		The number of branches.
»» pvGeneration	Double	kWh	The power produced of branch N.

5.3 Battery Storage

5.3.1 Querying the Latest Battery Data

5.3.1.1 API Overview

Queries the latest battery data based on the site ID or device SNs.

1. Site ID: Only one site can be queried at a time. All batteries at the site are queried.
2. Device SNs: Up to 100 batteries can be queried at a time.

Note: The site ID and device SNs are mutually exclusive. If the site ID is specified, device SNs are invalid.

5.3.1.2 Request URL

`https://xxxxxxxx/openapi/v1/battery/getBatteryLastData`

5.3.1.3 Request Method

HTTP method: GET

5.3.1.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteId	url	string	No	siteId and SNs are mutually exclusive. If siteId is specified, SNs is invalid. siteId : the site ID. Only one site ID is supported. SNs : the battery SNs. Up to 100 SNs are supported.
SNs	url	string	No	

5.3.1.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» siteId	string		The site ID.
»» SN	string		The battery SN.

»» lastReportedTime	Long		The data update time.
»» status	Integer		The battery status. Offline : offline. charge : charging. discharge : discharging. standby : standby. sleep : sleeping. fault : faulty. upgrading : upgrading. inverter_power_off : powered off.
»» power	Integer	W	The battery power.
»» batteryMode	string		The battery working mode. selfUse : self-powered. touMode : TOU.
»» SOC	Integer	%	
»» dailyCharging	double	kWh	The power daily charged.
»» dailyDischarge	double	kWh	The power daily discharged.
»» monthlyCharging	double	kWh	The power monthly charged.
»» monthlyDischarge	double	kWh	The power monthly discharged.
»» yearlyCharging	double	kWh	The power yearly charged.
»» yearlyDischarge	double	kWh	The power yearly discharged.
»» lifetimeCharging	double	kWh	The power cumulatively charged.
»» lifetimeDischarge	double	kWh	The power cumulatively discharged.

5.3.2 Querying Historical Battery Data

5.3.2.1 API Overview

Queries historical battery data based on the site ID or device SNs. By default, historical data of the current day is queried. To query data of another day, specify the date.

1. Site ID: Only one site can be queried at a time. All batteries at the site are queried.
2. Device SNs: Up to 100 batteries can be queried at a time.

Note: The site ID and device SNs are mutually exclusive. If the site ID is specified, device SNs are invalid.

5.3.2.2 Request URL

`https://xxxxxxxxx/openapi/v1/battery/getBatteryHistoryData`

5.3.2.3 Request Method

HTTP method: GET

5.3.2.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteId	url	string	No	siteId and SNs are mutually exclusive. If siteId is specified, SNs is invalid. siteId : the site ID. Only one site ID is supported. SNs : the battery SNs. Up to 100 SNs are supported.
SNs	url	string	No	
date	url	string	No	Format: dd/MM/yyyy. Default value: current day.

5.3.2.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» SN	string		The battery SN.
»» dateTime	String		Format: dd/MM/yyyy HH:mm. Interval: 15 minutes.
»» power	Integer	W	The active power.
»» SOC	Integer	%	

5.3.3 Querying Battery Power

5.3.3.1 API Overview

Queries historical battery power based on the site ID or device SNs. Data

can be queried by day, month, and year. By default, data of the current day is queried. To query data of another day, specify the date type and date.

1. Site ID: Only one site can be queried at a time. All batteries at the site are queried.
2. Device SNs: Up to 100 batteries can be queried at a time.

Limits: Data can be queried within 30 days or 12 months. If the time range to query exceeds a limit, status code 2007 is returned.

Note: The site ID and device SNs are mutually exclusive. If the site ID is specified, device SNs are invalid.

5.3.3.2 Request URL

`https://xxxxxxxxx/openapi/v1/battery/getBatterysDayData`

5.3.3.3 Request Method

HTTP method: GET

5.3.3.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteId	url	string	No	siteId and SNs are mutually exclusive. If siteId is specified, SNs is invalid. siteId : the site ID. Only one site ID is supported. SNs : the battery SNs. Up to 100 SNs are supported.
SNs	url	string	No	
dateType	url	string	No	The date type, such as day, month, or year. Default value: day.
beginDate	body	string	No	Day: dd/MM/yyyy. Month: MM/yyyy.

				Year: yyyy. Default value: current day.
endDate	body	string	No	Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy. Default value: current day, which is the same as beginDate .

5.3.3.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» SN	string		The battery SN.
»» dateType	string		The date type, such as day, month, or year.
»» date	string		Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy.
»» charging	double	kWh	The power charged.
»» discharge	double	kWh	The power discharged.

5.4 Gateway

5.4.1 Querying the Latest Gateway Data

5.4.1.1 API Overview

Queries the latest gateway data based on the site ID or device SNs.

1. Site ID: Only one site can be queried at a time. All gateways at the site are queried.
2. Device SNs: Up to 100 gateways can be queried at a time.

Note:

1. The site ID and device SNs are mutually exclusive. If the site ID is

specified, device SNs are invalid.

2. A single-phase gateway does not contain data of phase A, B, or C, and **null** is returned for phase data.

5.4.1.2 Request URL

`https://xxxxxxxxx/openapi/v1/gateway/getGatewaysLastData`

5.4.1.3 Request Method

HTTP method: GET

5.4.1.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteId	url	string	No	siteId and SNs are mutually exclusive. If siteId is specified, SNs is invalid. siteId : the site ID. Only one site ID is supported. SNs : the gateway SNs. Up to 100 SNs are supported.
SNs	url	string	No	

5.4.1.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» SN	string		The gateway SN.
»» lastReportedTime	Long		The data update time.
»» status	string		The gateway status. offline : offline. offlineFault : faulty and offline. normal : normal and online. fault : faulty and online. upgrading : upgrading.

»» solarGenerationPower	Integer	W	The solar active power.
»» solarGenerationPowerA	Integer	W	The solar active power of phase A.
»» solarGenerationPowerB	Integer	W	The solar active power of phase B.
»» solarGenerationPowerC	Integer	W	The solar active power of phase C.
»» solarReactivePower	Integer	W	The solar reactive power.
»» solarReactivePowerA	Integer	W	The solar reactive power of phase A.
»» solarReactivePowerB	Integer	W	The solar reactive power of phase B.
»» solarReactivePowerC	Integer	W	The solar reactive power of phase C.
»» dailySoalrGeneration	double	kWh	The solar power daily produced.
»» monthlySoalrGeneration	double	kWh	The solar power monthly produced.
»» yearlySoalrGeneration	double	kWh	The solar power yearly produced.
»» lifetimeSoalrGeneration	double	kWh	The solar power cumulatively produced.
»» gridPower	Integer	W	The grid power.
»» gridPowerA	Integer	W	The grid active power of phase A.
»» gridPowerB	Integer	W	The grid active power of phase B.
»» gridPowerC	Integer	W	The grid active power of phase C.
»» gridVoltage	Integer	V	The grid voltage.
»» gridVoltageA	Integer	V	The grid voltage of phase A.
»» gridVoltageB	Integer	V	The grid voltage of phase B.
»» gridVoltageC	Integer	V	The grid voltage of phase C.
»» dailyGridExport	double	kWh	The power daily exported.
»» dailyGridImport	double	kWh	The power daily imported.
»» monthlyGridExport	double	kWh	The power monthly exported.
»» monthlyGridImport	double	kWh	The power monthly imported.
»» yearlyGridExport	double	kWh	The power yearly exported.
»» yearlyGridImport	double	kWh	The power yearly imported.
»» lifetimeGridExport	double	kWh	The power cumulatively exported.
»» lifetimeGridImport	double	kWh	The power cumulatively imported.
»» dailyConsumption	double	kWh	The power daily consumed.
»» monthlyConsumption	double	kWh	The power monthly consumed.
»» yearlyConsumption	double	kWh	The power yearly consumed.
»» lifetimeConsumption	double	kWh	The power cumulatively consumed.
»» batteryPower	Integer	W	The battery power.
»» batteryPowerA	Integer	W	The battery active power of phase A.
»» batteryPowerB	Integer	W	The battery active power of phase B.
»» batteryPowerC	Integer	W	The battery active power of phase C.
»» batterySOC	double	%	The battery SOC.
»» batteryStatus	string		The battery status. charge : charging. discharge : discharging. normal : not charged or discharged.
»» batteryMode	string		The battery working mode.

			selfUse : self-powered. touMode : TOU.
»» dailyBatteryCharging	double	kWh	The battery power daily charged.
»» dailyBatteryDischarge	double	kWh	The battery power daily discharged.
»» monthly_batteryCharging	double	kWh	The battery power monthly charged.
»» monthlyBatteryCharging	double	kWh	The battery power monthly discharged.
»» yearlyBatteryCharging	double	kWh	The battery power yearly charged.
»» yearlyBatteryDischarge	double	kWh	The battery power yearly discharged.
»» lifetimeBatteryCharging	double	kWh	The battery power cumulatively charged.
»» lifetimeBatteryDischarge	double	kWh	The battery power cumulatively discharged.

5.4.2 Querying Historical Gateway Data

5.4.2.1 API Overview

Queries historical gateway data based on the site ID or device SNs. By default, historical data of the current day is queried. To query data of another day, specify the date.

1. Site ID: Only one site can be queried at a time. All gateways at the site are queried.
2. Device SNs: Up to 100 gateways can be queried at a time.

Note: The site ID and device SNs are mutually exclusive. If the site ID is specified, device SNs are invalid.

5.4.2.2 Request URL

<https://xxxxxxxxx/openapi/v1/gateway/getGatewaysHistoryData>

5.4.2.3 Request Method

HTTP method: GET

5.4.2.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteId	url	string	No	siteId and SNs are mutually exclusive. If siteId is specified, SNs is invalid. siteId : the site ID. Only one site ID is supported. SNs : the gateway SNs. Up to 100 SNs are supported.
SNs	url	string	No	
date	url	string	No	Format: dd/MM/yyyy. Default value: current day.

5.4.2.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» siteId	string		The site ID.
»» SN	string		The gateway SN.
»» dateTime	String		Format: dd/MM/yyyy HH:mm.
»»» solarGenerationPower	Integer	W	The solar active power.
»» solarReactivePower	Integer	W	The solar reactive power.
»» gridPower	Integer	W	The grid power.
»» gridVoltage	Integer	V	The grid voltage.
»» gridVoltageA	Integer	V	The grid voltage of phase A.
»» gridVoltageB	Integer	V	The grid voltage of phase B.
»» gridVoltageC	Integer	V	The grid voltage of phase C.
»» batteryPower	Integer	W	The battery power.
»» batterySOC	double	%	The battery SOC.

5.4.3 Querying Gateway Power

5.4.3.1 API Overview

Queries the historical power data of gateways based on the site ID or device SNs. Data can be queried by day, month, and year. By default,

data of the current day is queried. To query data of another day, specify the date type and date.

1. Site ID: Only one site can be queried at a time. All gateways at the site are queried.
2. Device SNs: Up to 100 gateways can be queried at a time.

Limits: Data can be queried within 30 days or 12 months. If the time range to query exceeds a limit, status code 2007 is returned.

Note: The site ID and device SNs are mutually exclusive. If the site ID is specified, device SNs are invalid.

5.4.3.2 Request URL

<https://xxxxxxxxx/openapi/v1/gateway/getGatewaysEnergy>

5.4.3.3 Request Method

HTTP method: GET

5.4.3.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	The session token.
siteId	url	string	No	siteId and SNs are mutually exclusive. If siteId is specified, SNs is invalid. siteId : the site ID. Only one site ID is supported. SNs : the gateway SNs. Up to 100 SNs are supported.
SNs	url	string	No	
» dateType	url	string	No	The date type, such as day, month, or year. Default value: day.
» beginDate	body	string	No	Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy. Default value: current day.

» endDate	body	string	No	Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy. Default value: current day, which is the same as beginDate .
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5.4.3.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» SN	string		The gateway SN.
»» dateType	string		The date type, such as day, month, or year.
»» date	string		Day: dd/MM/yyyy. Month: MM/yyyy. Year: yyyy.
»» soalrGeneration	double	kWh	The solar power produced.
»» consumption	double	kWh	The power consumed.
»» gridExport	double	kWh	The power exported.
»» gridImport	double	kWh	The power imported.
»»batteryCharging	double	kWh	The battery power charged.
»»batteryDischarge	double	kWh	The battery power discharged.

5.5 Device Alerts

5.5.1 Querying Active Device Alerts

5.5.1.1 API Overview

Queries active device alerts based on the site ID or device SNs. Alerts can be queried by time or level. Up to 100 alerts can be returned at a time. If the number of alerts exceeds 100, another call is required, and pagination is used. If **pageEnd=1** is returned, all alerts have been returned.

1. Site ID: Up to 100 sites can be queried at a time.
2. Device SNs: Up to 100 devices can be queried at a time.

Note: The site ID and device SNs are mutually exclusive. If the site ID is specified, device SNs are invalid.

5.5.1.2 Request URL

`https://xxxxxxxx/openapi/v1/device/getAlarmsBySite`

5.5.1.3 Request Method

HTTP method: GET

5.5.1.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	
Steck-Accept-Language	header	string	No	For more information, see Languages in the appendix. Default: en-US .
siteId	url	string	No	siteId : the site IDs. Up to 100 site IDs are supported.
SNs	url	string	No	SNs : the device SNs. Up to 100 SNs are supported.
beginDate	url	Long	No	Format: in milliseconds. Default value: all.
endDate	body	Long	No	Format: in milliseconds. Default value: all, which is the same as beginDate .
level	body	String	No	The alert level, such as critical, major, minor, or warning. Default value: all.
page	url	integer	No	The page number. Default value: 1 .

5.5.1.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was

			successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»» siteId	string		The site ID.
»» deviceSN	string		The device SN.
»» alarmId	string		The alert ID.
»» alarmName	string		The alert name.
»» alarmLevel	string		The alert level.
»» alarmOccurTime	string		The alert occurrence time.
»» alarmCauseID	string		The cause ID.
»» alarmCause	string		The cause.
»»alarmRepairSuggestion	string		The recommendations.
» pageEnd	Integer		The pagination status. 0 : incomplete. 1 : complete.

6 Control APIs

Control APIs allow you to query and adjust solar, battery, and grid parameters. These APIs are applicable only to sites with one gateway.

6.1 Sending Control Tasks

6.1.1 API Overview

Sets and controls solar, battery, and grid parameters based on the site ID.

Up to 100 sites are supported at a time. For more information, see Control Parameters in the appendix.

6.1.2 Request URL

<https://xxxxxxxx/openapi/v1/device/paramReading>

6.1.3 Request Method

HTTP method: POST

6.1.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	
Body	body	object	Yes	
»SNs	body	List	Yes	The gateway SNs. Up to 100 gateway SNs are supported.
»»SN	body	String	Yes	The gateway SN.
»»»params	body	List	Yes	The parameters.
»»»» paramCode	body	String	Yes	The parameter name.

6.1.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	object		
»» requestId	string		The task ID.
»» results	List		The returned results.
»»» siteId	string		The site ID.
»»»» SN	string		The gateway SN.
»»»» subTaskId	string		The subtask ID, which is the globally unique identifier allocated to each gateway in the control task. It can be used to query the task execution results of the gateway.

6.2 Querying Control Task Results

6.2.1 API Overview

Queries the parameter execution status based on the task ID (**requestId**) or subtask IDs (**subTaskIds**). For more information, see Control

Parameters in the appendix.

1. Task ID: Only one task can be queried at a time. The execution status of all parameters in the task is queried.
2. Subtask IDs: Up to 100 subtasks can be queried at a time. The execution status of the parameters in each subtask is queried.

Note: The task ID and subtask IDs are mutually exclusive. If the task ID is specified, subtask IDs are invalid.

6.2.2 Request URL

`https://xxxxxxxxx/openapi/v1/device/getParamSettingTask`

6.2.3 Request Method

HTTP method: GET

6.2.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	
Body	body	object	Yes	
»requestId	body	String	Yes	requestId and subTaskIds are mutually exclusive. If requestId is specified, subTaskIds is invalid. requestId : the task ID. Only one task ID is supported at a time. subTaskIds : the subtask IDs. Up to 100 subtask IDs are supported at a time.
»subTaskIds	body	List	No	
»» subTaskId	body	string	No	The subtask ID.

6.2.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful.

			false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	list		
»»requestID	string		The task ID.
»»subtaskId	List		The subtask ID.
»»siteid	string		The site ID.
»»SN	string		The gateway SN.
»»paramResults	List		The parameter execution results.
»»» paramCode	String		The parameter name.
»»»paramValue	String		The parameter value.
»»»PSitStatus	Integer		The parameter execution status. 0 : pending. 1 : successful. 2 : failed.
»»»PSitTime	String		The execution time.

6.3 Sending Parameter Query Tasks

6.3.1 API Overview

Queries solar, battery, and grid parameters based on the site ID. Up to 100 sites are supported at a time. It is recommended to call this API before a control task is sent or after a control task is executed successfully. For more information, see Control Parameters in the appendix.

6.3.2 Request URL

<https://xxxxxxxxx/openapi/v1/device/paramRead>

6.3.3 Request Method

HTTP method: POST

6.3.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	
Body	body	object	Yes	
»SNs	body	List	Yes	The gateway SNs. Up to 100 gateway SNs are supported.
»»SN	body	String	Yes	The gateway SN.
»»»params	body	List	Yes	The parameters.
»»»»paramCode	body	String	Yes	The parameter name.

6.3.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	object		
»» requestID	string		The task ID.
»» results	List		The returned results.
»»» siteId	string		The site ID.
»»» SN	string		The gateway SN.
»»»» subTaskID	string		The subtask ID, which is the globally unique identifier allocated to each gateway in the query task. It can be used to query the task query results of the gateway.

6.4 Querying Parameter Query Task Results

6.4.1 API Overview

Queries the parameter execution status based on the task ID or subtask IDs. For more information, see Control Parameters in the appendix.

1. Task ID: Only one task can be queried at a time. The execution status of all parameters in the task is queried.
2. Subtask IDs: Up to 100 subtasks can be queried at a time. The

execution status of the parameters in each subtask is queried.

Note: The task ID and subtask IDs are mutually exclusive. If the task ID is specified, subtask IDs are invalid.

6.4.2 Request URL

https://xxxxxxxxx/openapi/v1/device/getParamReadingTask

6.4.3 Request Method

HTTP method: GET

6.4.4 Request Parameters

Parameter	Position	Type	Required	Description
Authorization	header	string	Yes	
Body	body	object	Yes	
»requestID	body	String	No	requestID and subTaskIDs are mutually exclusive. If requestID is specified, subTaskIDs is invalid. requestID : the task ID. Only one task ID is supported at a time. subTaskIDs : the subtask IDs. Up to 100 subtask IDs are supported at a time.
»subTaskIDs	body	List	No	
»»subTaskID	body	List	No	The subtask ID.

6.4.5 Response Parameters

Parameter	Type	Unit	Description
» success	boolean		true : The request was successful. false : The request failed.
» code	integer		These parameters are returned if the value of success is false . For more information, see Status Codes in the appendix.
» reason	string		
» details	string		
» data	listobject		
»»RequestID	string		The task ID.
»»subtaskID	string		The returned results.
»»siteid	string		The site ID.
»»SN	string		The gateway SN.

»»paramResults	List		The parameter execution results.
»»»paramCode	String		The parameter name.
»»»PReadStatus	integer		The parameter query status. 0: pending. 1: successful. 2: failed.
»»»PReadTime	String		The query time.
»»»paramValue	String		The parameter value.

7 Appendixes

7.1 Status Codes

No.	Status Code	Cause	Description
1	200	success	success
2	400	parameter validation failed	The input has an empty value or a nonstandard format
3	500	system error	Server system exception
4	2004	the siteId cannot be empty	In the required parameter, The siteId cannot be empty
5	2005	the siteIds cannot be empty	In the required parameter, The siteIds cannot be empty
6	2006	Incorrect date format	The entered date format is not transmitted according to the standard format. Procedure
7	2007	The time range query exceeds the limit	Each API has a corresponding time query range, please query according to the time range
8	2402	The number of third-party systems exceeds the limit	
9	2403	The third-party system has expired	
10	2404	The key does not exist or has been deleted The system has been disabled or the key is unavailable	
11	2405	The key has expired or has expired	
11	2406	The system name already exists	

7.2 Languages

No.	Language Code	Description
1	de-DE	German
2	en-US	English
3	es-ES	Espana
4	fr-FR	French
5	it-IT	Italy
6	pl-PL	Poland
7	th-TH	Thailand
8	zh-CN	China

7.3 Country Codes

No.	GEC Country Code	STANAG 1059 Country Code	Description
1	AF	AFG	Afghanistan
2	AL	ALB	Albania
3	AG	DZA	Algeria
4	AQ	ASM	American Samoa
5	AN	AND	Andorra
6	AO	AGO	Angola
7	AV	AIA	Anguilla
8	AY	ATA	Antarctica
9	AC	ATG	Antigua and Barbuda
10	AR	ARG	Argentina
11	AM	ARM	Armenia
12	AA	ABW	Aruba
13	AS	AUS	Australia
14	AU	AUT	Austria
15	AJ	AZE	Azerbaijan
16	BF	BHS	Bahamas
17	BA	BHR	Bahrain
18	BG	BGD	Bangladesh
19	BB	BRB	Barbados
20	BO	BLR	Belarus
21	BE	BEL	Belgium
22	BH	BLZ	Belize
23	BN	BEN	Benin
24	BD	BMU	Bermuda
25	BT	BTN	Bhutan
26	BL	BOL	Bolivia

27	BK	BIH	Bosnia and Herzegovina
28	BC	BWA	Botswana
29	BV	BVT	Bouvet Island
30	BR	BRA	Brazil
31	IO	IOT	British Indian Ocean Territory
32	VI	VGB	British Virgin Islands
33	BX	BRN	Brunei
34	BU	BGR	Bulgaria
35	UV	BFA	Burkina Faso
36	BM	MMR	Myanmar
37	BY	BDI	Burundi
38	CV	CPV	Cape Verde
39	CB	KHM	Cambodia
40	CM	CMR	Cameroon
41	CA	CAN	Canada
42	CJ	CYM	Cayman Islands
43	CT	CAF	Central African Republic
44	CD	TCD	Chad
45	CI	CHL	Chile
46	CH	CHN	Chinese mainland
47	KT	CXR	Christmas Island
48	CK	AUS	Cocos (Keeling) Islands
49	CO	COL	Colombia
50	CN	COM	Comoros
51	CG	COD	The Democratic Republic of the Congo
52	CF	COG	The Republic of the Congo
53	CW	COK	Cook Islands
54	CS	CRI	Costa Rica
55	IV	CIV	Côte d'Ivoire
56	HR	HRV	Croatia
57	CU	CUB	Cuba
58	UC		Curaçao
59	CY	CYP	Cyprus
60	EZ	CZE	The Czech Republic
61	DA	DNK	Denmark
62	DJ	DJI	Djibouti
63	DO	DMA	Dominica
64	DR	DOM	Dominican Republic
65	EC	ECU	Ecuador
66	EG	EGY	Egypt
67	ES	SLV	El Salvador
68	EK	GNQ	Equatorial Guinea

69	ER	ERI	Eritrea
70	EN	EST	Estonia
71	ET	ETH	Ethiopia
72	FK	FLK	Falkland Islands
73	FO	FRO	Faroe Islands
74	FJ	FJI	Fiji
75	FI	FIN	Finland
76	FR	FRA	France
77	FG	GUF	French Guiana
78	FP	PYF	French Polynesia
79	FS	ATF	French Southern Territories
80	GB	GAB	Gabon
81	GA	GMB	Gambia
82	GZ	PSE	Palestine
83	GG	GEO	Georgia
84	GM	DEU	Germany
85	GH	GHA	Ghana
86	GI	GIB	Gibraltar
87	GR	GRC	Greece
88	GL	GRL	Greenland
89	GJ	GRD	Grenada
90	GP	GLP	Guadeloupe
91	GQ	GUM	Guam
92	GT	GTM	Guatemala
93	GK		Guernsey
94	GV	GIN	Guinea
95	PU	GNB	Guinea Bissau
96	GY	GUY	Guyana
97	HA	HTI	Haiti
98	HM	HMD	Heard Island and McDonald Islands
99	VT	VAT	Vatican City
100	HO	HND	Honduras
101	HK	HKG	Hong Kong (China)
102	HU	HUN	Hungary
103	IC	ISL	Iceland
104	IN	IND	India
105	ID	IDN	Indonesia
106	IR	IRN	Iran
107	IZ	IRQ	Iraq
108	EI	IRL	Ireland
109	IM		Isle of Man
110	IS	ISR	Israel
111	IT	ITA	Italy

112	JM	JAM	Jamaica
113	JA	JPN	Japan
114	JE		Jersey
115	JO	JOR	Jordan
116	KZ	KAZ	Kazakhstan
117	KE	KEN	Kenya
118	KR	KIR	The Republic of Kiribati
119	KN	PRK	DPRK
120	KS	KOR	Republic of Korea
121	KV		Kosovo
122	KU	KWT	Kuwait
123	KG	KGZ	Kyrgyzstan
124	LA	LAO	Laos
125	LG	LVA	Latvia
126	LE	LBN	Lebanon
127	LT	LSO	Lesotho
128	LI	LBR	Liberia
129	LY	LBY	Libya
130	LS	LIE	Liechtenstein
131	LH	LTU	Lithuania
132	LU	LUX	Luxembourg
133	MC	MAC	Macao (China)
134	MK	FYR	North Macedonia
135	MA	MDG	Madagascar
136	MI	MWI	Malawi
137	MY	MYS	Malaysia
138	MV	MDV	Maldives
139	ML	MLI	Mali
140	MT	MLT	Malta
141	RM	MHL	Marshall Islands
142	MB	MTQ	Martinique
143	MR	MRT	Mauritania
144	MP	MUS	Mauritius
145	MF	FRA	Mayotte
146	MX	MEX	Mexico
147	FM	FSM	Micronesia
148	MD	MDA	Moldova
149	MN	MCO	Monaco
150	MG	MNG	Mongolia
151	MJ	MNE	Montenegro
152	MH	MSR	Montserrat
153	MO	MAR	Morocco
154	MZ	MOZ	Mozambique

155	WA	NAM	Namibia
156	NR	NRU	Nauru
157	NP	NPL	Nepal
158	NL	NLD	Netherlands
159	NC	NCL	New Caledonia
160	NZ	NZL	New Zealand
161	NU	NIC	Nicaragua
162	NG	NER	Niger
163	NI	NGA	Nigeria
164	NE	NIU	Niue
165	NF	NFK	Norfolk Island
166	CQ	MNP	Northern Mariana Islands
167	NO	NOR	Norway
168	MU	OMN	Oman
169	PK	PAK	Pakistan
170	PS	PLW	Palau
171	PM	PAN	Panama
172	PP	PNG	Papua New Guinea
173	PA	PRY	Paraguay
174	PE	PER	Peru
175	RP	PHL	The Philippines
176	PC	PCN	Pitcairn Islands
177	PL	POL	Poland
178	PO	PRT	Portugal
179	RQ	PRI	Puerto Rico
180	QA	QAT	Qatar
181	RE	REU	Réunion
182	RO	ROU	Romania
183	TW	TWN	Taiwan (China)
184	RS	RUS	Russia
185	RW	RWA	Rwanda
186	TB		Saint Barthélemy
187	SH	SHN	Saint Helena, Ascension and Tristan da Cunha
188	SC	KNA	Saint Kitts and Nevis
189	ST	LCA	Saint Lucia
190	RN		Saint Martin
191	VC	VCT	Saint Vincent and the Grenadines
192	WS	WSM	Samoa
193	SM	SMR	San Marino
194	TP	STP	São Tomé and Príncipe
195	SA	SAU	Saudi Arabia
196	SG	SEN	Senegal

197	RI	SRB	Serbia
198	SE	SYC	Seychelles
199	SL	SLE	Sierra Leone
200	SN	SGP	Singapore
201	NN		Sint Maarten
202	LO	SVK	Slovakia
203	SI	SVN	Slovenia
204	BP	SLB	Solomon Islands
205	SO	SOM	Somalia
206	SF	ZAF	South Africa
207	SX	SGS	South Georgia and the South Sandwich Islands
208	OD	SSD	South Sudan
209	SP	ESP	Spain
210	CE	LKA	Sri Lanka
211	SU	SDN	Sudan
212	NS	SUR	Suriname
213	SB	SPM	Saint Pierre and Miquelon
214	WZ	SWZ	Eswatini
215	SW	SWE	Sweden
216	SZ	CHE	Switzerland
217	SY	SYR	Syria
218	TI	TJK	Tajikistan
219	TZ	TZA	Tanzania
220	TH	THA	Thailand
221	TT	TLS	Timor-Leste
222	TO	TGO	Togo
223	TL	TKL	Tokelau
224	TN	TON	Tonga
225	TD	TTO	Trinidad and Tobago
226	TS	TUN	Tunisia
227	TU	TUR	Türkiye
228	TX	TKM	Turkmenistan
229	TK	TCA	Turks and Caicos Islands
230	TV	TUV	Tuvalu
231	UG	UGA	Uganda
232	UP	UKR	Ukraine
233	AE	ARE	United Arab Emirates
234	UK	GBR	United Kingdom
235	US	USA	United States of America
236	UM		United States Minor Outlying Islands
237	UY	URY	Uruguay

238	UZ	UZB	Uzbekistan
239	NH	VUT	Vanuatu
240	VE	VEN	Venezuela
241	VM	VNM	Viet Nam
242	VQ	VIR	U.S. Virgin Islands
243	WF	WLF	Wallis and Futuna
244	WI	ESH	Western Sahara
245	YM	YEM	Yemen
246	ZA	ZMB	Zambia
247	ZI	ZWE	Zimbabwe

7.4 Device Types

No.	Device Type	Enumeration
0	System	system
1	Site	station
2	Gateway	gateway
3	General power generating device	generator
4	Inverter	inverter
5	Microinverter	micro_inverter
6	General energy consuming device	consumer
7	Battery storage	storage
8	Component	component
9	Meter	meter
10	Heat pump	heat_pump
11	Smart device	smart_device

7.5 Control Parameters

Parameter	Description	Unit	Read / Write	Value Range
Battery parameters				
endOfChargeSOC	The end-of-charge SOC.	%	Read and write	Range: [80,100]. Precision: two decimal places.
endOfDischargeSOC	The end-of-discharge SOC.	%	Read and write	Range: [0,30]. Precision: two decimal places.
chargeMaxPower	The maximum charging power.	kW	Read and write	The maximum charging power per battery.

				Range: [0,Maximum rated charging power of a battery]. Precision: two decimal places. You can call the <code>getDevicesBySite</code> API to query the maximum rated charging power of a battery.
<code>dischargeMaxPower</code>	The maximum discharging power.	kW	Read and write	The maximum discharging power per battery. Range: [0,Maximum rated discharging power of a battery]. Precision: two decimal places. You can call the <code>getDevicesBySite</code> API to query the maximum rated discharging power of a battery.
<code>batteryMode</code>	The battery working mode.		Read and write	1: self-powered. 2: TOU.
<code>batteryControl</code>	The forced charge or discharge mode.		Read and write	0: forced charge. 1: forced discharge. 2: exits forced charge or discharge. (The battery will enter the specified working mode.) Forced charge or discharge does not change the original working mode of the battery. If the charge or discharge task ends, the battery continues to work in the

				original mode.
batteryControlMode	The forced charge or discharge method.		Read and write	0 : based on the target SOC. 1 : based on the charge or discharge duration. This parameter is required if forced charge or discharge is enabled.
batteryControlTargetSOC	The target SOC of forced charge or discharge.	%	Read and write	Range: [0,100]. Precision: two decimal places. This parameter is required if batteryControlMode is set to 0 .
batteryControlTimes	The forced charge or discharge duration.	Minutes	Read and write	Range: [0,1440]. Precision: zero decimal places. This parameter is required if batteryControlMode is set to 1 .
batteryControlPower	The forced charging or discharging power.	kW	Read and write	Charging power range: [0,SUM(Maximum charging power of all batteries)]. Discharging power range: [0,SUM(Maximum discharging power of all batteries)]. Precision: two decimal places.
Solar parameters				
solarPowerMode	The control mode of active power.			0 : no limit. 2 : percentage control.
solarPowerPercentage	The scheduling percentage of active power.	%		Range: [0,100]. Precision: one decimal place.
solarReactiveMode	The control			0 : no limit.

	mode of reactive power.			3 : power factor (PF).
solarReactivePF	The PF.			Range: [-1,-0.8] U [0.8,1]. Precision: three decimal places.
System parameters (They are of the highest priority. If they are specified, battery and solar parameters are subject to their values.)				
gridPowerLimit	The grid connection with limited power.			0 : disabled. 1 : enabled.
gridPowerLimitMode	The power limit method.			0 : total power. 1 : single-phase power.
gridPowerLimitMax	The maximum power to grid.	kW		Range: [0,1000]. Precision: three decimal places.